Inflation relative price variance

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1 Introduction

Inflation is known to have adverse welfare effects through its distortion of relative prices which causes the misallocation of resources, production inefficiencies leading to lower real economic activity. This is well supported by both theoretical and empirical studies. This paper contributes to the empirical literature, by examining the relationship between inflation and relative price variability (RPV) in South Africa given the little evidence available and to better understand how this relationship has evolved and the implications for monetary policy.

The South African context can provide further insights into the relationship between inflation and RPV given the shifts in the inflation targets from 3-6 per cent in 2000 to 4.5 per cent since 2017 as structural changes in inflation can affect the link between inflation and relative price variability (Caglayan and Filiztekin, 2003) and that the functional form of relationship is U-shaped and dynamic (Fielding and Mizen, 2000; Choi, 2010). The latter motivates the estimation of the inflation rate that minimizes relative price distortions since if the relationship was linear then the optimal inflation rate would be zero. Given the inflation target reform towards a 3% in South Africa, this study contributes to the policy discussion on the appropriate inflation target and understating how dynamic relations between inflation and relative price variability has evolved.

The theoretical literature generally distinguishes between menu-cost and signal extraction models in explaining the positive relation between trend inflation and RPV. In these models' firms follow a one-sided dynamic pricing rule when setting prices and face fixed non-zero costs of adjusting nominal prices (e.g. printing new menus) when inflation increases as the latter reduces the optimal real price of goods and services. Due to differences in administrative costs, these nominal price adjustment costs vary across firms moreover these adjustments occur at different times which implies staggard price setting behaviour which leads to relative price variability. Menu cost models predict a that higher expected inflation causes an increase in RPV (citations xxx) while Lucas-type signal extraction models (Lucas(1973); Hercowitz (1981); Cukierman (1984) xxxx more citatiosn) argue unexpected inflation uncertainty causes higher RPV because informational frictions make it difficult to distinguish between idiosyncratic relative shocks or an aggregate (inflationary) shock. Menu cost and signal-extraction models assume a positive and linear relationship making the optimal inflation rate zero, however, recently monetary search models (Head and Kumar, 2005) show that this relationship is nonlinear making the optimal inflation rate somewhere between zero and positive inflation rate.

2 Citations

Adam et al. (2024) Baglan et al. (2016) Ball and Romer (1993) Benabou and Gertner (1993) Caglayan and Filiztekin (2003) Caglayan et al. (2008) Grier and Perry (1996) Lach and Tsiddon (1992) Nakamura et al. (2018) Ndou and Redford (2014) Ndou and Gumata (2017) Parks (1978) Parsley (1996) Rather et al. (2018)

3 Literature review

4 Data and methodology

Our analysis uses monthly inflation data on 45 goods categories from January 2009 to March 2025 obtained from Statistics South Africa (Stats SA).[1] **The data is seasonally adjusted using ???.** Figure A1 below shows the long term inflation rate in South Africa, and Figure A2 in the appendix shows the various price categories used in the analysis. Table A1 provides the descriptive statistics of the inflation rates for the various price categories used in the analysis.

The relative price variance (RPV) is calculated in line with Lach and Tsiddon (1992) as the weighted standard deviation of individual price inflation rates from the aggregate inflation rate. The RPV is essentially the variability of relative prices of a given product. We first calculate the monthly inflation rate as follows:

$$\pi_{i,t} = ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) * 100, \tag{1}$$

[1]The data is available at: http://www.statssa.gov.za/

where $\pi_{i,t}$ is the inflation rate of good i at time t, $P_{i,t}$ is the price of good i at time t, and $P_{i,t-1}$ is the price of good i at time $t-1$. Then, the RPV is calculated as:

$$RPV_t = \sqrt{\sum_{i=1}^{n} \omega_i (\pi_{i,t} - \pi_t)^2}, \tag{2}$$

where ω_j is the expenditure share of the j^{th} good, π_t is the headline or aggregate inflation rate at time t, and $\sum_{i=1}^{n} \omega_i = 1$. The expenditure weights (ω_i) used are derived from the Consumer Price Index (CPI) basket provided by Stats SA. Figure 1 shows the RPV calculated using the above formula.**When where these weigths constructed???**. The weights are shown in Figure A3 in the appendix.

To examine the relationship between inflation and RPV, we estimate the following econometric model:

$$RPV_t = \beta_0 + \beta_1 |\pi_t| + \sum_{h=1}^{3} \omega_h RPV_{t-h} + \mu_t, \tag{3}$$

where RPV_t is the relative price variance at time t, $|\pi_t|$ is the absolute value of aggregate inflation rate at time t, RPV_{t-h} are lagged values of RPV to account for its dynamic nature, and μ_t is the error term. The coefficient β_1 captures the effect of inflation on RPV. We include three lags of RPV based on Akaike Information Criterion (AIC) (and other tests) and to capture potential persistence in RPV. For robustness, we repeat this estimation using core inflation (excluding food and non-alcoholic beverages, and fuel and electricity) instead of headline inflation to account for the potential volatility in these components.

The literature on RPV emphasises that the relationship between inflation and RPV maybe time-varying due to structural changes in the economy, monetary policy regimes, and other factors. To capture potential time variation in the relationship, we also estimate a rolling regression model with a **24-month and 60-month windows** as follows:

$$RPV_t = \alpha_t + \beta_{1,t} |\pi_t| + \sum_{h=1}^{3} \omega_{h,t} RPV_{t-h} + \mu_t, \tag{4}$$

where the coefficients α_t, $\beta_{1,t}$, and $\omega_{h,t}$ vary over time. This approach allows us to observe the stability of the coefficients over time, which is an indicator of non-linearity in the relationship between RPV and inflation. Furthermore, we employ a quantile on quantile regression approach, which is robust to outliers and allows us to regime aspects of the relationship between RPV and inflation. For brevity, we explain the quantile on quantile estimation procedure in the Appendix.

Lastly, we utilise a semi-parametric or partially linear approach to estimate the optimal level of inflation given the RPV. This approach allows us to model the relationship between inflation and RPV without imposing a strict functional form. Following the literature, we first specify a quadratic (U-shaped) relationship between inflation and RPV as follows:

$$RPV_t = \alpha + \pi_t + \beta_2 \pi_t^2 + \sum_{h=1}^{3} \phi_h RPV_{t-h} + \epsilon_t, \tag{5}$$

where β_2 captures the curvature of the relationship between inflation and RPV. The optimal inflation rate that minimizes RPV can be derived by taking the first derivative of the equation with respect to π_t and setting it to zero, yielding:

$$\pi_t^* = -\frac{1}{2\beta_2}. \tag{6}$$

To estimate this optiomal inflation we implement a semi parametric approach as follows:

$$RPV_t = f(\pi_t) + \sum_{h=1}^{3} \phi_h RPV_{t-h} + \epsilon_t, \tag{7}$$

where $f(\pi_t)$ is a non-linear smooth differentiable function of inflation and the RPV. Therefore, the optimal inflation rate can be obtained by finding the minimum of the estimated $f(\pi_t)$ function. $f(\pi_t)$ is estimated in two stages. First we estimate λ_h as follows:

$$RPV_t = \sum_{h=1}^{3} \lambda_h \overline{RPV}_{t-h} + \eta_t, \tag{8}$$

where $\overline{RPV}_{t-h}$ are the residuals from a non-parametric regression of each lag of RPV_t on π_t. In the second stage, we estimate $f(\pi_t)$ non-parametrically as follows:

$$\hat{\eta}_t = f(\pi_t) + v_t, \tag{9}$$

where $\hat{\eta}_t$ are the residuals from Equation 8, and $f(\pi_t)$ is estimated using a kernel regression method. In both stages we use the Nadaraya-Watson kernel estimator. The bandwidth (h) for the kernel estimator is selected using cross-validation to minimize the mean squared error. As a treatment for extreme values of inflation, we further implement the unbounded Gaussian kernel and the outlier-robust Epanechnikov kernel. In essence, the semi-parametric approach captures the sensitivity of RPV to marginal increases in inflation. In that case, the optimal inflation rate can be obtained by finding the point where the first derivative ($f'(\pi_t)$) is equal to zero. Therefore, if $f'(\pi_t) > 0$ ($f'(\pi_t) < 0$), then an increase in inflation (decrease) leads to an increase (decrease) in the RPV.

5 Results

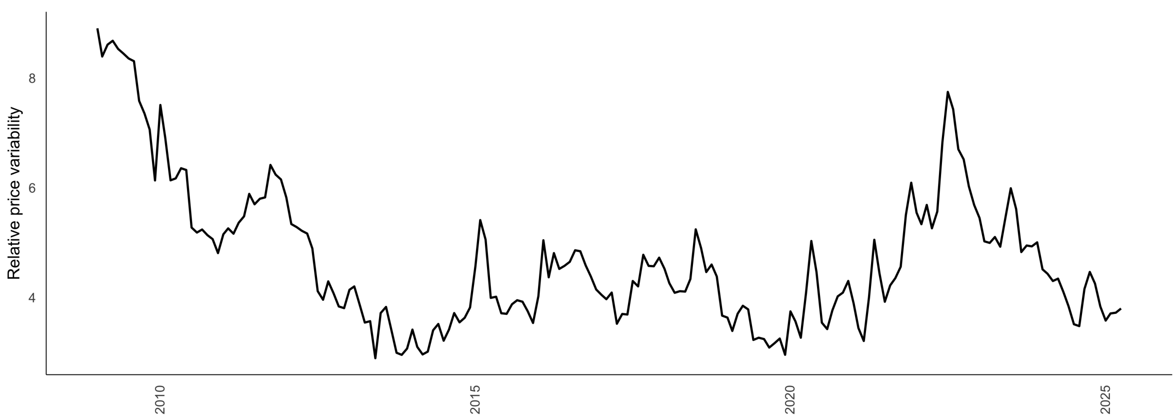

Figure 1: RPV over-time

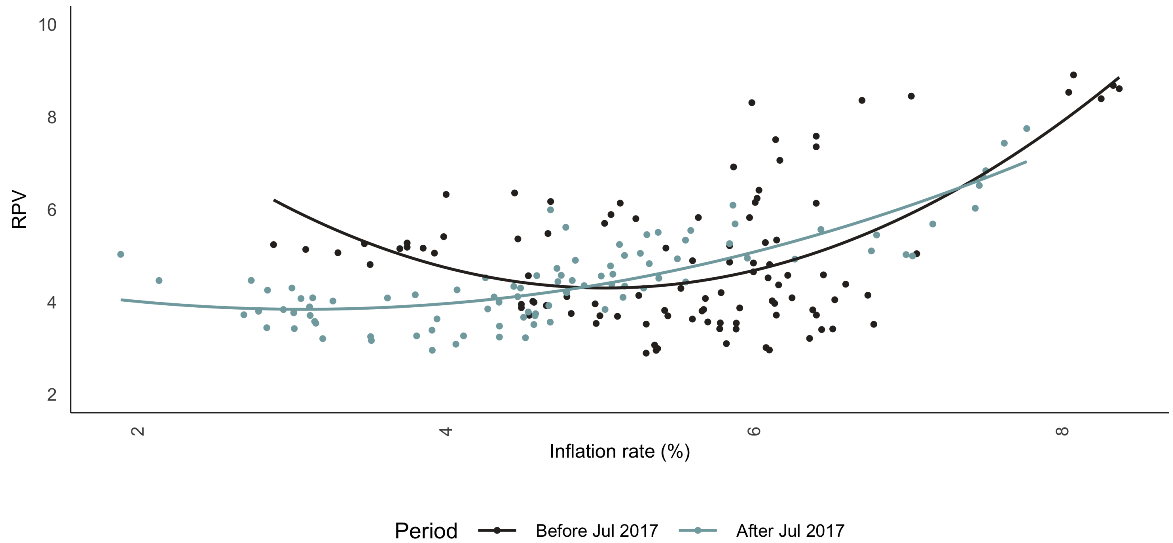

Figure 2: RPV vs headline Inflation

Table 1: Baseline regressions

	Full Sample	Pre 2017	Post 2017	Full Sample: Squared Inflation	Pre 2017: Squared Inflation	Post 2017: Squared Inflation
$\|\pi_t\|$	0.009*	0.009	0.032***			
π_t				-0.047	-0.104	-0.031
π_t^2				0.006*	0.011*	0.007
RPV_{t-3}	0.247***	0.173*	0.259**	0.237***	0.154	0.223**
RPV_{t-2}	-0.386***	-0.14	-0.66***	-0.367***	-0.111	-0.626***
RPV_{t-1}	1.039***	0.896***	1.12***	1.017***	0.864***	1.071***
D_{2022m8}	0.027		0.026	0.001		-0.008
D_{2020m2}	-0.116		-0.215**	-0.107		-0.208**
D_{2017m7}	0.156*			0.161*		
D_{2014m7}	-0.097	-0.087		-0.104	-0.096	
$D_{2011m12}$	0.042	0.027		0.047	0.037	
C	0.097**	0.049	0.265***	0.248**	0.375*	0.473***

Note: *** = p value < 0.01, ** = 0.01 < p value < 0.05, * = 0.05 < p value < 0.1.

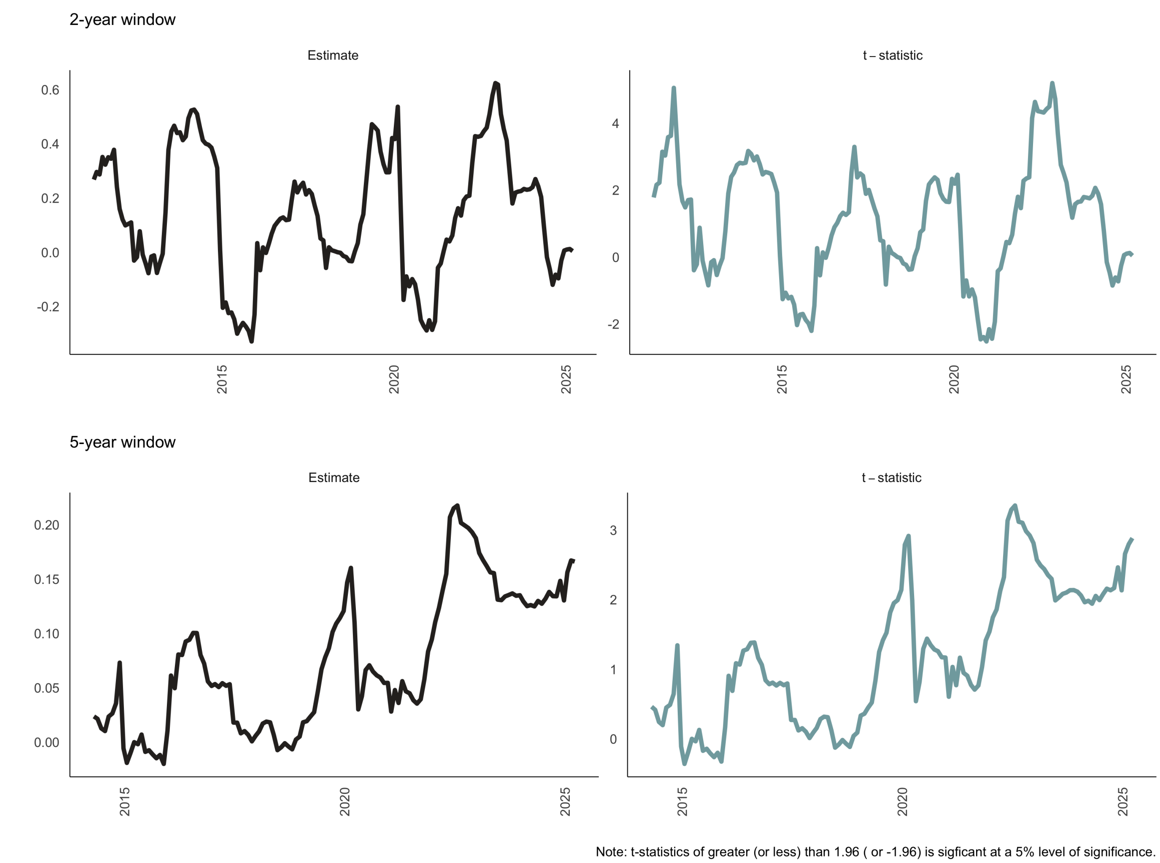

Figure 3: Rolling regressions

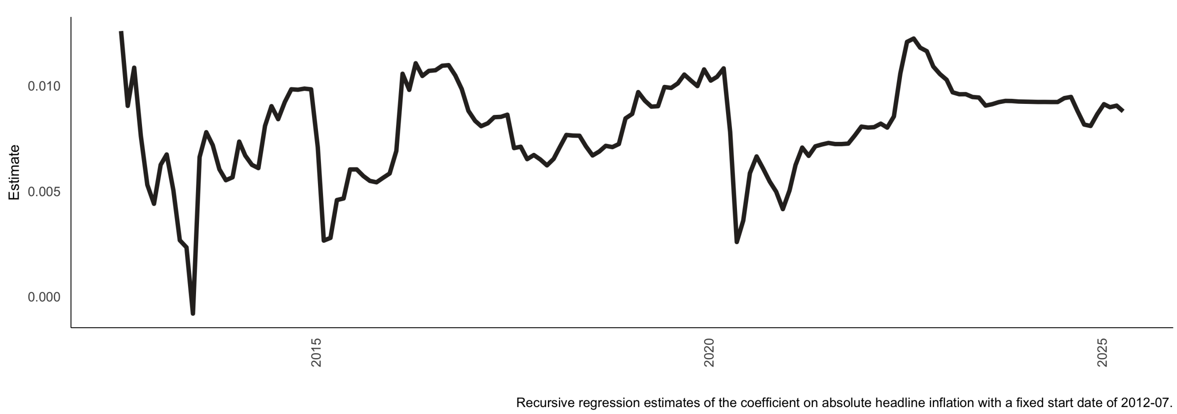

Figure 4: Recursive regressions

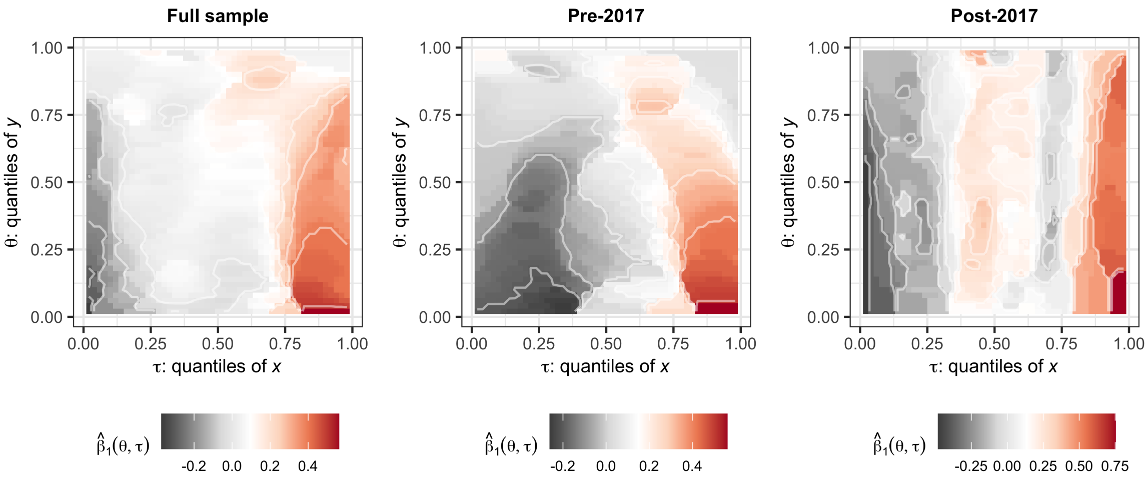

Figure 5: Quantile on quantile regressions

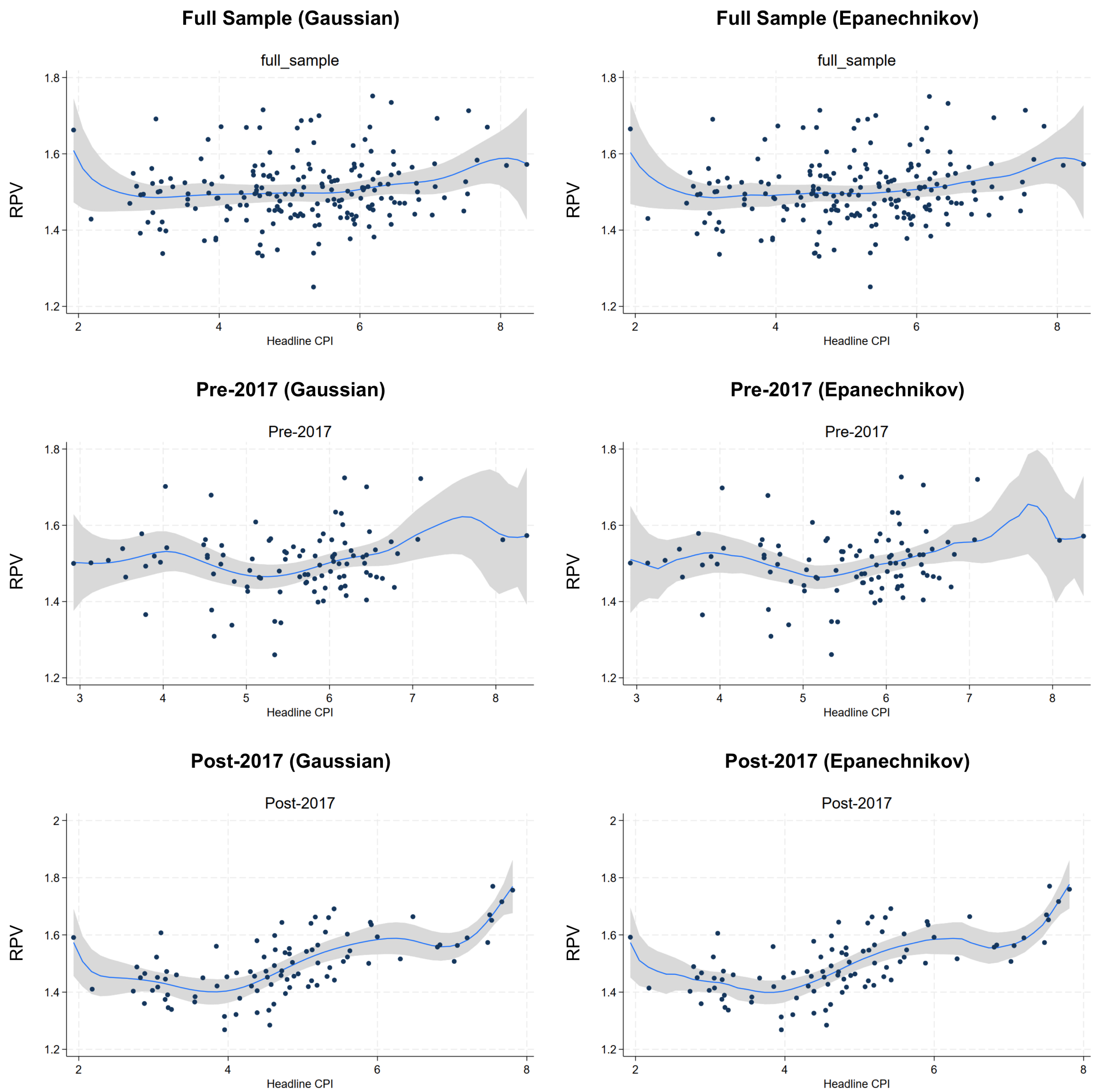

Figure 6: Semi-parametric regressions

6 Conclusion

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A Appendix: Data and descriptive statistics

A.1 Data

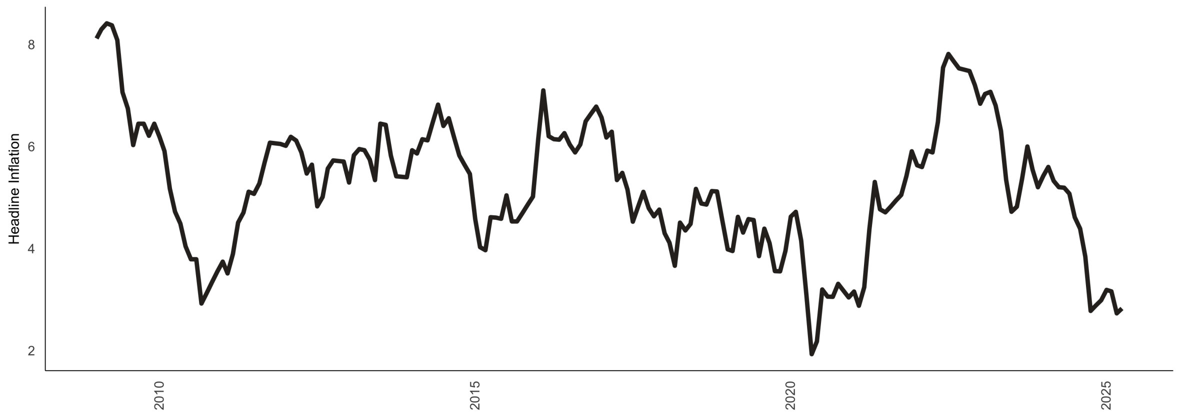

Figure A1: Headline inflation (%).

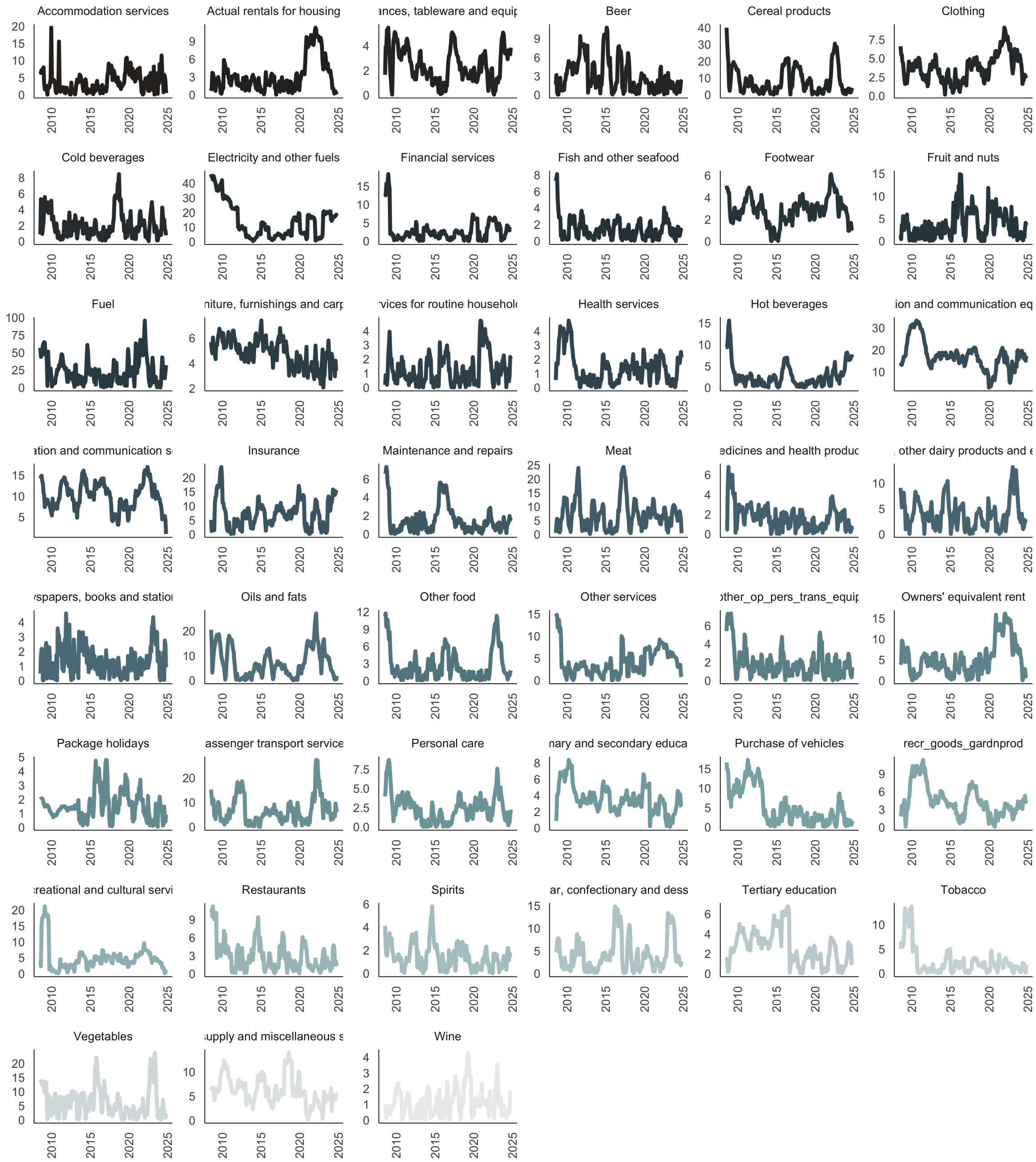

Figure A2: Inflation categories (%)

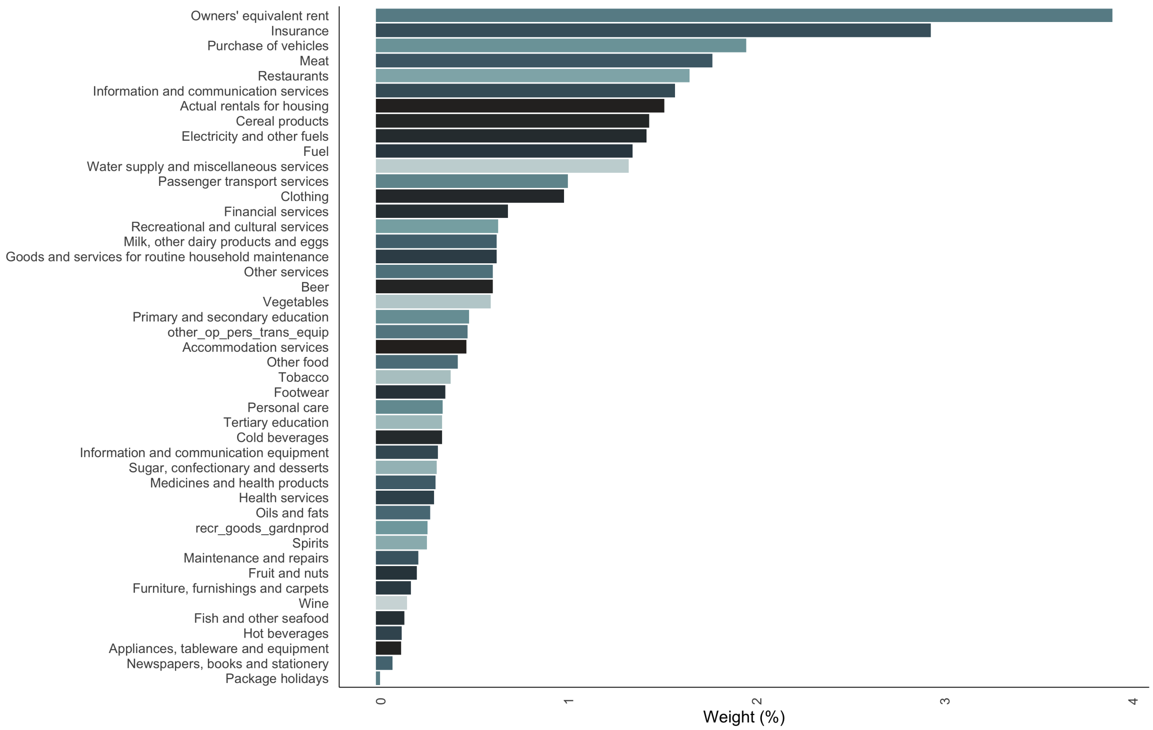

Figure A3: CPI Weights. Note: Due to a lack of data, some categories (no spitirits; transport services of goods; household textiles; glassware,tableware, and house utensils; and Tools and equipment for house and garden) have been excluded.

A.2 Descriptive statistics

Table A1: Descriptive statistics

Goods category	Mean	Median	SD	Min	Max	Observations
Accommodation services	3.69	2.97	2.96	0.02	19.60	196
Actual rentals for housing	3.19	2.39	2.73	0.04	11.28	196
Appliances, tableware and equipment	2.40	2.13	1.35	0.03	5.47	196
Beer	2.97	2.04	2.53	0.01	10.77	196
Cereal products	8.33	4.98	7.98	0.01	40.01	196
Clothing	3.74	3.61	1.78	0.17	9.09	196
Cold beverages	2.01	1.62	1.57	0.04	8.45	196
Electricity and other fuels	13.98	10.03	11.77	0.53	46.31	196
Financial services	2.76	1.97	3.14	0.02	18.30	196
Fish and other seafood	1.46	1.15	1.24	0.01	8.09	196
Footwear	2.84	2.87	1.22	0.03	6.13	196
Fruit and nuts	3.69	2.99	3.07	0.02	14.90	196
Fuel	22.89	18.12	18.11	0.13	95.60	196
Furniture, furnishings and carpets	4.77	4.80	1.08	2.07	7.39	196
Goods and services for routine household maintenance	1.25	0.97	1.05	0.00	4.71	196
Health services	1.37	1.24	1.10	0.00	4.75	196
Hot beverages	2.74	1.94	2.75	0.03	15.61	196
Information and communication equipment	17.26	16.87	6.42	2.86	33.39	196
Information and communication services	10.04	10.12	3.25	0.91	16.99	196
Insurance	7.27	7.10	4.85	0.12	23.75	196
Maintenance and repairs	1.51	1.09	1.53	0.01	7.29	196
Meat	7.12	6.30	5.30	0.15	24.32	196
Medicines and health products	1.72	1.54	1.26	0.01	6.79	196
Milk, other dairy products and eggs	3.99	3.64	2.81	0.05	13.17	196
Newspapers, books and stationery	1.31	1.11	1.00	0.02	4.59	196
Oils and fats	7.39	6.15	5.99	0.00	27.17	196
Other food	2.67	1.70	2.74	0.06	11.73	196
Other services	4.15	3.19	3.07	0.00	15.10	196
Owners' equivalent rent	5.29	4.75	3.98	0.01	16.18	196
Package holidays	1.55	1.40	1.00	0.04	4.80	196
Passenger transport services	6.74	5.40	5.36	0.00	27.30	196
Personal care	2.44	2.33	1.77	0.01	8.72	196
Primary and secondary education	3.60	3.39	1.69	0.13	8.37	196
Purchase of vehicles	5.22	3.33	4.67	0.08	17.30	196
Recreational and cultural services	4.75	4.05	3.89	0.07	21.16	196
Restaurants	2.91	2.29	2.42	0.07	11.17	196
Spirits	1.43	1.30	1.04	0.00	5.80	196
Sugar, confectionary and desserts	4.18	3.18	3.75	0.03	14.87	196
Tertiary education	2.61	2.50	1.71	0.02	6.78	196
Tobacco	2.41	1.57	2.89	0.01	13.75	196
Vegetables	6.35	5.05	5.12	0.00	23.81	196
Water supply and miscellaneous services	6.16	5.77	2.83	0.15	13.91	196
Wine	1.13	0.98	0.83	0.03	4.28	196
other_op_pers_trans_equip	1.85	1.44	1.58	0.00	7.33	196
recr_goods_gardnprod	4.39	3.86	2.56	0.06	11.34	196

A.3 Stationarity tests

Table A2: Stationarity tests results

Measure	ADF test (p-value)	PP test (p-value)	KPSS test (p-value)	Stationarity
Headline Inflation	0.19	0.24	0.08	Stationary
RPV	0.09	0.34	0.01	Stationary

A.4 Lag length selection

Table A3: RPV lag length selection

Tests	Optimal Lags
AIC(n)	3
HQ(n)	3
SC(n)	1
FPE(n)	3

B Appendix: Alternative Results (OLS Robustness)

The core RPV (excluding food and non-alcoholic beverages, and fuel and electricity) is calculated using the same formula as in Equation 2 but using the core inflation rate instead of the headline inflation rate. The results are consistent with the main findings, indicating a positive relationship between core inflation and core RPV.

B.1 Core RPV over-time

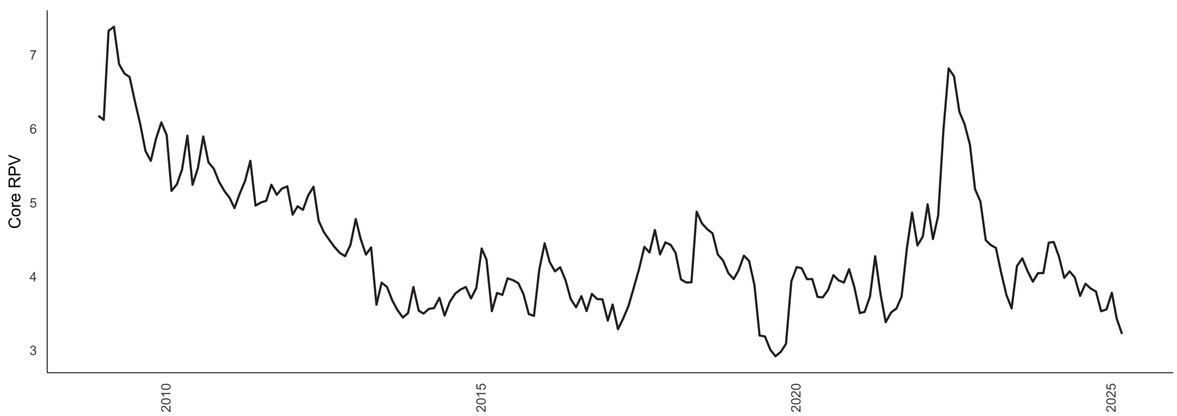

Figure B1: Core RPV overtime

B.2 Stationarity tests for core RPV

Table B1: Stationarity tests results

Measure	ADF test (p-value)	PP test (p-value)	KPSS test (p-value)	Stationarity
Core Inflation	0.02	0.62	0.01	Stationary
Core RPV	0.10	0.23	0.01	Stationary

B.3 Lag length selection for core RPV

Table B2: Core RPV lag length selection

Tests	Optimal Lags
AIC(n)	1
HQ(n)	1
SC(n)	1
FPE(n)	1

B.4 Core RPV baseline regression results

Table B3: Core RPV baseline regressions

	Full Sample	Pre 2017	Post 2017	Full Sample: Squared Inflation	Pre 2017: Squared Inflation	Post 2017: Squared Inflation
$\lvert\pi_t\rvert$	0.003	0.003	0.01			
π_t				-0.034	-0.106**	0.165
π_t^2				0.004	0.01**	-0.02
RPV_{t-1}	0.924***	0.946***	0.865***	0.909***	0.866***	0.875***
D_{2022m8}	0.023*		0.05	0.034		0.041
D_{2020m2}	0.047***		0.046	0.046		0.037
D_{2017m7}	0.055***			0.057		
D_{2014m7}	-0.078***	-0.074		-0.077	-0.076	
$D_{2011m12}$	0.035***	0.032		0.038	0.034	
$Constant$	0.092**	0.06	0.148*	0.201**	0.46***	-0.161

Note: *** = p value < 0.01, ** = 0.01 < p value < 0.05, * = 0.05 < p value < 0.1.

C Appendix: Coefficient of variation version of RPV

The coefficient of variation (CV) version of RPV (or RPV-CV) normalizes the RPV by the aggregate price level to account for changes in the overall price level. This measure is particularly useful when comparing RPV across different time periods or economies with varying inflation rates. The RPV-CV is calculated as follows:

$$RPV_t = \sqrt{\frac{\sum_{i=1}^{n} \omega_i (\pi_{i,t} - \pi_t)^2}{|1 + \pi_t|}}, \tag{10}$$

where π_t is the aggregate inflation rate at time t, and ω_i are the inflation basket weights.

D Appendix: Explanation of the quantile on quantile regression

To further capture the dynamic dependencies in the data, we perform a quantile on quantile regressions. Quantile regressions address the limitations of OLS which only estimates the mean dependency between on or more independent variables and the dependant variable. Distribution moments, such as the mean, can be strongly affected by heavy tails (or tail behaviour), non-linearity and extreme values in general (see Koenker (2017)). In response to these limitations, the quantile regression was first introduced by Koenker and Bassett (1978). Koenker (2017) notes that quantile regressions are inherently local and are immune to small deviations in distributions.

However, Gupta et al. (2018) note that standard quantile regressions are limited in their ability to capture dependency in its entirety. That is, although quantile regressions capture the relationship between two variable (for example) at various points of the conditional distribution, it restricts the possibility that the nature of the independent variable can also influence how the independent variable is calculated. The quantile-on-quantile regression (QQR), therefore, offers a more complete picture of the complex dependency structure, and was utilised numerous times in the literature (see Mishra et al. (2019), Chang et al. (2020), and Sim (2016)).

Given this brief explanation on QQR, we estimate the relationship between the θ-quantile of RPV (a_{2t}) and the τ-quantile of headline inflation (Γ_t) as follows:

$$a_{2t} = \beta^{\theta}\Gamma_t + \epsilon_t^{\theta}, \tag{11}$$

Taking the linear version of $\beta^{\theta}(.)$ with a first order Taylor expansion of $\beta^{\theta}(.)$ around Γ^{τ} gives:

$$\beta^{\theta}(\Gamma_t) \approx \beta^{\theta}(\Gamma^{\tau}) + \beta^{\theta'}(\Gamma^{\tau})(\Gamma_t - \Gamma^{\tau}) \tag{12}$$

In this instance both $\beta^{\theta}(\Gamma^{\tau})$ and $\beta^{\theta'}(\Gamma^{\tau})$ are indexed to θ and τ. This means that Equation 12 can be summarised as follows:

$$\beta^{\theta}(\Gamma_t) \approx \beta_0(\theta, \tau) + \beta_1(\theta, \tau)(\Gamma_t - \Gamma), \tag{13}$$

and collecting terms gives and substituting into Equation 11 results in :

$$a_{2t} = \beta_0(\theta, \tau) + \beta_1(\theta, \tau)(\Gamma^{\tau})(\Gamma_t - \Gamma) + \epsilon_t^{\theta} \tag{14}$$

where $\beta^{\theta}(\Gamma^{\tau})$ and $\beta^{\theta'}(\Gamma^{\tau})$ are substituted with $\beta_0(\theta, \tau)$ and $\beta_1(\theta, \tau)(\Gamma^{\tau})$ respectively.

Therefore, Equation 14 captures the relationship between the θ-quantile of the RPV and the τ-quantile of inflation, that is, the overall dependence structure of the respective distributions. To this end, to get the estimates of $\hat{\beta}_0(\theta, \tau)$ and $\hat{\beta}_1(\theta, \tau)$ we solve for:

$$\underset{b_0, b_1}{min} \sum_{i=1}^{n} \rho_{\theta} \left[a_{2t} - b_0 - b_1(\widehat{\Gamma_t} - \hat{\Gamma}) \right] K \left(\frac{F_n(\widehat{\Gamma_t}) - \tau}{h} \right) \tag{15}$$

where ρ_{θ} is a tilted absolute value function which produces θ-quantile values of a_{2t}. A Gaussian kernel $K(.)$ is used to weight observations in the neighbourhood of $\widehat{\Gamma^{\tau}}$ using h as a bandwidth. These weights are inversely related to $\widehat{\Gamma_t} - \hat{\Gamma}$. Lastly, the choice of h remains uncertain in kernel regression where if a small h is chosen the bias of the estimates is smaller but the variance of these estimates increases, and conversely. To overcome this uncertainty we use the cross validation method to calculate the appropripate level of h.